

Ethical Hacking

Notes collected during lessons. Not corrected.

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IMPORTANT DISCLAIMER: These notes have been taken during lessons. They might contain errors (specifically, grammatical errors) and be incorrect. Please always check other sources to confirm what is written is correct. I don't think I'll correct them after lessons, though it's possible.

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Chapter 1

Introduction

1.1 Course Introduction

For the recommended textbook, we are following
Hacking Exposed 7: Network Security Secrets & Solutions.

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The major topics we'll cover are:

- Footprinting
- Scanning
- Enumeration
- Advanced Persistent Threat
- Hacking:
 - Web Hacking
 - Unix Hacking
 - Wireless Hacking
 - Hardware Hacking
 - Mobile Hacking

It's recommended to train with HackTheBox website. If you get +45 points on the website, you are awarded +1 at the final grade. Another website that is recommended is given by Try to Hack (or something like this).

We'll have a Lab project. It's optional, and it will account for 49% of the final grade. The rest of the grade will be given by a written exam. If we want to skip the lab project, then 100% of the final grade will be given by the final exam. The written exam is given by 5 questions.

Chapter 2

Part 1

We'll start with taking a look at **Footprinting, Scanning and Enumeration**.

Attackers, before performing an attack, will perform some kind of investigation to determine the vulnerabilities they can use at their own advantage. We are considering professional hackers, who don't want to be detected when they're entering the system.

Attackers behave as insiders when breaking a system, even if he could be outside the company. When we perform Ethical Hacking, we need to be inside the company, having the same vision an hacker has.

Also, the major concern of an hacker is to be anonymous, as well as being stealthy. At the basic of anonymity there is **Tor Project**. As the graphical user interface for Tor, Vidalia is used, while for web filtering proxy we have Privoxy. To solve IP addresses, it's possible to do this Tor as well, with **tor-resolve**. For proxychains, **Nmap** is well known. It's a tool that allows to do some sophisticated protocols. Once another tool is given by **nc** (netcat), used to send requests to servers.

Onion Router (TOR)

We have the usual scheme, with Alice wanting to talk with Bob. However, they want the possibility to deny they're communicating, leaving no trace of their communication. The protocol works with three layers of routers, located around the world, and there are hundreds of thousand of them. Alice will pick three random routers. These routers will have special names: **guard node**, the node to which Alice is talking (entry point of the network); **exit-node**. This is called Onion Router because the packet that is used is composed by many envelopes. The main message is covered in many other messages. We have both encrypted and unencrypted links. Encryption is implemented by Tor protocol, but on top of this we can have more encryption on top of it. Tor protocol is also transparent to the communication. The packet being sent is encrypted three times, with three different keys, one after the other. The key is given by the Tor node that Alice has selected. Each node has a public and a private key. The guard node is the most external envelope of the encryption. The guard node then decrypts the outside of the message, forwarding it to the second node.

So the sender prepares these onion messages, with every node removing one layer, and passing it to the next of the three nodes (randomly picked nodes).
Correct this part, I got a bit lost.

2.1 Footprinting

So far, we've seen tools to find information about the target we attack. Before that, however, Footprinting is performed. It works without even sending to the target a single packet. So, we need to find information about the organization without even interacting. Footprinting should be executed also by us Ethical Hackers. This is internet Footprinting:

1. Determine the scope of your activities: we need to understand what I actually need to do, avoiding to inadvertently committing crimes.
2. Get proper authorization: the scope should be linked with our authorization to the system. The contract will tell us what we can and what we cannot do.
3. Check publicly available information
4. WHOIS & DNS enumeration
5. DNS interrogation
6. Network Reconnaissance